Nicholas Tang

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Portfolio — GitHub — LinkedIn

Education

UC Santa Cruz Sept 2024 – Jun 2027
B.S. Computer Science GPA: 3.9
Relevant Coursework: Signals and Systems, Electronic Circuits, Embedded OS, Computer Architecture, Data Structures

Experience

Rivian and Volkswagen Group Technologies June 2026 – Sept 2026
Software Engineering Intern, Charging
- Extended protocol conformance for DIN 70121 EV charging communications, expanding unit and Software-in-the-Loop (SIL) test suites to permanently validate and lock in protocol compliance.
- Automated Hardware-in-the-Loop (HIL) integration tests in Python on a general-purpose communications test bench to evaluate cross-brand Rivian and VW vehicle software.
- Configured and implemented 2 new Unified Diagnostic Services (UDS) Data Identifiers (DIDs) on charging ECUs to enforce State of Charge (SoC) limits and enable dynamic pluglock control.
- Conducted hands-on in-vehicle testing to validate UDS DIDs on target vehicle hardware and participated in system requirement discussions to refine diagnostic functionality.

Formula Slug (FSAE) @ UC Santa Cruz Sept 2024 – Present
Firmware Engineer
- Implemented real-time CAN-bus lap tracking firmware in C++ using flat-Earth position approximation, maintaining spatial accuracy within a ±10-meter margin across thousands of telemetry data points.
- Built on VN-200 IMU/GPS module SDK, enabling asynchronous data streaming for deterministic vehicle dynamics data acquisition.
- Developed STM32 MbedOS firmware for closed-loop PID traction control and regenerative braking energy management to improve vehicle handling stability and energy efficiency.
- Engineered safety-critical low-voltage protection routines featuring rapid undervoltage detection and automated relay actuation to protect battery cell integrity.

UCSC Baskin Engineering (ECE) Mar 2026 – Present
Undergraduate Researcher, High Performance Power Electronics Lab
- Designed a multi-layer breakout PCB in Altium Designer for the TI C2000 microcontroller family to accelerate lab prototyping.
- Integrated dual synchronous buck converters (5V → 3.3V and 3.3V → 1.2V) to deliver stable rail voltages for MCU core and I/O.
- Executed PCB layout routing, trace optimization, ground plane design, and bill of materials (BOM) sourcing.

UCSC Earth and Planetary Sciences Jun 2025 – Jan 2026
Undergraduate Researcher, Planetary Cloud Tracking
- Developed C/C++ computer vision pipelines implementing optical flow algorithms to extract planetary atmospheric wind vectors from satellite imagery.
- Formulated numerical integrators to solve continuous Euler-Lagrange fluid equations to model atmospheric motion.

Projects & Other Experience

NASA's Professional Development Program (NPWEE) June 2025 – Aug 2025
Lead Systems Engineer
- Led systems architecture design and co-authored a technical proposal for electric aircraft mid-air battery swap infrastructure, conducting trade studies on airspace integration and automated power transfer.

Musical Auto-Transcribe DSP Dec 2025 – Present
Personal Project
- Engineered a C-based digital signal processing pipeline that converts raw audio into human-readable musical notation and MIDI output.
- Applied Fast Fourier Transforms (FFT), Hann windowing, and custom denoising filters to perform spectral analysis and pitch detection on complex audio signals.

Skills

Languages: C, C++, Python, Bash
Embedded & HW: RTOS (MbedOS), STM32, TI C2000, CAN bus, UART, SPI, UDS Diagnostics, HIL/SIL Testing
EDA & Tools: Altium Designer, KiCad, Siemens NX, Oscilloscopes, CMake, Ninja, Docker, Valgrind, GDB, Git